MILITARY SPECIFICATION

DELAY ELEMENT M2, FOR FUZES M557 AND M524 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing for one type of delay element designated as Delay Element M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1169 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9258182	-	Box, Set-Up, Packing, Ammunition, Inner, for Small Explosive Components
9258183	-	Box, Set-Up, Packing, Ammunition, Outer, for Small Explosive Components
9258184	-	Box, Packing, Ammunition, for Small Explosive Components
9276524	-	Delay Element, M2, Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET-9276524	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Assembly.-The delay element shall comply with all requirements specified on Drawing (dwg.) 9276524, and with all requirements specified in applicable specifications.

3.2 Moisture Content.-The moisture content of the following explosives at the loading station at the time of loading the delay elements shall not exceed the percentage specified in Table I.

TABLE I

Explosive	Percentage	Test Procedure
Primer Mixture NOL #130	0.30 maximum (max.)	4.4.1.1
Lead Azide	0.50 max.	4.4.1.2
Delay Composition	0.80 max.	4.4.1.3

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3.3 The delay element shall comply with the following:

3.3.1 Time.-The delay element shall function high order on the first attempt under an applied energy of 15.68 inch-ounces, within the time limits specified in Table II, when tested as specified in 4.4.3.1.

TABLE II

Delay Time (Sec.)	Lower Limit (Sec.)	Upper Limit (Sec.)
0.025	0.002	0.048
0.050	0.025	0.065

3.3.2 Delay Elements that Fail to Fire on the First Attempt (Duds).-The delay element shall not dud when tested as specified in 4.4.3.1.

3.4 Waterproofness.-The delay element shall comply with the ambient requirements of 3.3.1 and shall not dud after being submerged in water to a minimum (min.) depth of two (2) inches for not less than 48 hours, at a temperature of 70 degrees (°) Fahrenheit (F) plus or minus (±) 10°F, when tested as specified in 4.4.3.2.

3.5 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings for the moisture content of each explosive shall be submitted with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation as specified herein. The first article sample shall consist of inert parts (See 4.2.2) and loaded delay elements (See 4.2.3) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inert Parts.-A sample of twenty-five (25) each of every part shall be submitted in accordance with the provisions of 4.2.1, for inspection as specified in 4.2.2.1.

4.2.2.1 Inspections to be Performed.-The parts will be subjected by the Government to any or all of the applicable examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings.

4.2.2.2 Rejection.-If any part fails to comply with any of the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon the failure of an item to comply with any of the requirements.

4.2.3 Loaded Delay Elements.-The contractor shall submit a sample of forty-two (42) delay element assemblies to the Government for examination as specified in 4.3.2.5, and testing as specified in 4.2.3.1.

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4.2.3.1 Tests to be Performed.-Thirty-two (32) delay element assemblies shall be fired to determine time and duds, and ten (10) delay elements shall be water-immersion conditioned and fired for time and dud determinations.

4.2.3.2 Rejection.-The loaded first article sample shall be rejected if any delay element fails to comply with a requirement. In the event of rejection, the Government reserves the right to terminate its inspection upon any failure of a delay element to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded delay elements shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.

b. Primer mix NOL #130 or delay composition.-Each explosive from not more than one batch or blend of batches which results in homogeneity of each explosive used in loading a lot.

c. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision, and which have subsequently been subjected to uniform controlled drying and operating conditions at the loading plant.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-12 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (See dwg. 9276525, covering a detail of dwg. 9276524).

Categories	Defects	Method of Inspection	Code No. (See 6.2
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Inside diameter, max.	Gage	01001
102.	Depth of charge cavity, max.	Gage	01002
103.	Runout of hole to pitch diameter	Gage	01003
104.	Runout of outside diameter to pitch diameter	Gage	01004
105.	Pitch diameter, min.	Gage	01005
106.	Major diameter, min.	Gage	01006
Minor:	AQL 2.50 Percent		
201.	Length, min.	Gage	01007
202.	Outside diameter, min.	Gage	01008
203.	Hole diameter, max.	Gage	01009
204.	Width of slot, max.	Gage	01010
205.	Evidence of poor workmanship (See 3.5)	Visual	01011

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4.3.2.2 Disc, Closing (See dwg. 9276534, covering a detail of dwg. 9276524).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Diameter	Gage	02001
102.	Thickness	Gage	02002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	02003

4.3.2.3 Cup, Primer (See dwg. 9276527, covering a detail of dwg. 9276524).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	True position of coined surface to outside diameter	Gage	03001
102.	Thickness through coined surface	Gage	03002
Minor:	AQL 1.50 Percent		
201.	Length	Gage	03003
202.	Outside diameter	Gage	03004
203.	Evidence of poor workmanship (See 3.5)	Visual	03005

4.3.2.4 Primer Assembly (See dwg. 9276526, covering a detail of dwg. 9276534).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split, cut or cracked	Visual	04001

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Major:	AQL 0.40 Percent		
101.	Outside diameter, max.	Gage	04002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	04003

4.3.2.5 Assembly (See dwg. 9276524).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Presence of explosive on outside of body	Visual	05001
2.	Split, cut or crack in body	Visual	05002
Major:	AQL 0.65 Percent		
101.	Length, max.	Gage	05003
102.	Diameter, max.	Gage	05004
103.	Lacquer seal missing either end	Visual	05005
104.	Crimp not full 360°	Visual	05006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	05007

4.3.2.6 Box, Set-Up, Inner, After Loading and Prior to Closing Slide Cover (See dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Filler & cushion assembly missing	Visual	06001
2.	Filler & cushion assembly improperly assembled	Visual	06002
3.	Stiffener missing	Visual	06003
4.	Slide cover assembled reversed	Visual	06004
Major:	AQL 0.40 Percent		
101.	Release paper missing	Visual	06005
Minor:	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	06006
202.	Evidence of poor workmanship (See 3.5)	Visual	06007

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4.3.2.7 Box, Set-Up, Outer, with Inner Set-Up Boxes Assembled and Prior to Assembling Top (See dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Red surfaces on inner set-up boxes do not alternate on same sides of outer set-up boxes	Visual	07001
	2. Stiffener not facing one direction	Visual	07002
Major:	None defined.		
Minor:	AQL 1.00 Percent		
	201. Marking missing, incomplete or illegible	Visual	07003
	202. Evidence of poor workmanship (See 3.5)	Visual	07004

4.3.2.8 Box, Packing, Ammunition, Prior to Assembling Cover and Strapping (See dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
	101. Bags containing set-up boxes damaged	Visual	08001
	102. Cushioning material incomplete or missing from around outer set-up boxes	Visual	08002
Minor:	AQL 0.65 Percent		
	201. Evidence of poor workmanship (See 3.5)	Visual	08003

4.3.2.9 Box, Packing, Ammunition (See dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 Percent		
101.	Strapping missing, broken, loose or improperly fastened.....	Manual-Visual	09001
102.	Box damaged.....	Visual	09002
103.	Marking missing, broken.....	Visual	09003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5).....	Visual	09004

4.3.3 Testing.-Unless otherwise specified herein, the following tests shall be performed using MIL-STD-105 for sampling plan requirements.

4.3.3.1 Moisture Content (See 3.2, Table I) - Major Defects

TABLE III

Explosive	Code No.
Primer Mixture NOL 130	05008
Lead Azide	05009
Delay Composition	05010

The contractor shall provide adequate controls to insure that the explosives comply with the requirements. For verification, the contractor shall select a sample for test from that quantity of each explosive used in each eight (8) hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of explosive represented by the sample shall be rejected. If components have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1.

4.3.3.2 Composition of Primer Mixture NOL 130 (See dwg. 9276533, covering a detail of dwg. 9276524) - Major Defect - Code No. 05011.-The composition of each batch or blend of batches shall be determined at time of loading. A sample of sufficient size to perform the applicable test specified in 4.4.2 shall be selected. If the composition of the sample fails to comply with the drawing requirements, the quantity of explosive represented by the sample shall be rejected. Any detonator loaded with the non-conforming explosive shall be rejected.

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4.3.3.3 Functioning (See 3.3).-The delay element shall be observed for compliance with the requirements as classified in Table II. Both "Delay Time and Duds" shall be determined from the same sample. The test shall be performed as specified in 4.4.3.1.

TABLE IV

	Defect	Classification	Code No.
(1)	Duds	Major	10001
(2)	Delay Time	Major	10002

4.3.3.3.1 Duds.-The sample plan for this test shall be in accordance with MIL-STD-105, Single Sampling Plan and an AQL of 0.15 percent.

4.3.3.3.2 Delay Time.-The sample plan for this test shall be in accordance with MIL-STD-105, Single Sampling Plan and an AQL of 0.65 percent.

4.3.3.4 Waterproofness (See 3.4) - Major Defect - Code No. 10003.-A sample of 35 delay elements shall be selected at random from each lot. If one or more delay elements fails to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.3.3.2.

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-9276524.

4.3.4.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.4 Test Methods and Procedures

4.4.1 Moisture Content

4.4.1.1 Primer Mixture NOL 130.-The moisture content of the primer mixture shall be determined in accordance with Method 101.6 of MIL-STD-650.

4.4.1.2 Lead Azide.-The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650.

4.4.1.3 Delay Composition.-The moisture content of the delay composition shall be determined in accordance with Method 101.5 of MIL-STD-650.

4.4.2 Composition of Primer Mixture NOL 130 shall be determined as follows:

4.4.2.1 Primer Mixture NOL 130

4.4.2.1 Barium Nitrate.-An accurately weighed portion of approximately 0.5 gram (g) of primer mixture shall be transferred to a tared, medium porosity, fritted glass crucible of 20 milliliters (ml) capacity. Three (3) ml of lead azide saturated distilled water shall be added at a temperature of 50° Centigrade (C), ± 2°C, agitated for one (1) minute and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five (5) times, making a total of six (6) 3-ml extractions. The sample in the filtering crucible shall be rinsed three (3) times with alcohol, dried in an oven maintained at 60°C ± 5°C for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium Nitrate, Percent} = \frac{M\,(A-B)\,100}{1-0.01M}$$

Where:

A = Weight of crucible and sample

B = Weight of crucible and sample after extractions with lead azide saturated water

M = Moisture content of sample obtained from 4.4.1.1 in percent

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4.4.2.1.2 Basic Lead Styphnate (When Applicable).-The residue obtained in the determination of barium nitrate shall be extracted with six (6) ml portions of saturated ammonium acetate solution at about 25°C (hot saturated ammonium acetate shall not be used, as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml aliquot of this solution shall be transferred to a 50-ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm), and its absorbence density determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbence density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbence density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic Lead Styphnate, Percent} = \frac{F\,(A-B)}{W\,(1-0.01M)\,D}$$

Where:

A = Optical density of basic lead styphnate solution
B = Optical density of ammonium acetate solution
D = Width of corex cell, cm
M = Moisture content of sample obtained from 4.4.1.1 in percent
W = Weight of sample
F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard

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4.4.2.1.3 Determination of Normal Lead Styphnate (When Applicable).-Extract the residue obtained in the barium nitrate determination with sufficient 5-ml portions of ammonium acetate solution (saturated) at about 25°C until the washings are colorless Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for 1 additional minute. Finally, wash the residue with water containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbence density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution and measure at a wave length of 410 millimicrons, the absorbence density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal Lead Styphnate, Percent} = \frac{F\,(A-B)}{WD}$$

Where:

A = Optical density of normal lead styphnate solution
B = Optical density of ammonium acetate solution
D = Width of corex cell, cm
W = Weight of the dried sample
F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard

4.4.2.1.4 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three (3) times with alcohol, dried in an oven maintained at 60°C ± 5°C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml beaker with a stream of water; 25 ml of water shall be added and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three (3) times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60°C ± 5°C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

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$$\text{Tetracene, Percent} = \frac{(C-D)\ 100}{W\ (1-0.01M)}$$

Where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution
D = Weight of crucible and residue after treating with hot water
W = Weight of sample
M = Moisture content of sample obtained from 4.4.1.1 in percent

4.4.2.1.5 Antimony Sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony Sulfide, Percent} = \frac{(D-E)\ 100}{W\ (1-0.01M)}$$

Where:

D = Weight of crucible and sample after treating with hot water
E = Weight of crucible
W = Weight of sample
M = Moisture content of sample obtained in 4.4.1.1 in percent

4.4.2.1.6 Lead Azide.-The lead azide content in the sample, on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

4.4.3 Functioning

4.4.3.1 Delay Time and Duds.-The delay element shall be assembled to the applicable test fixture specified in 4.3.4. A steel ball 15/16 inch in diameter and weighing 1.96 ounces, max., shall be dropped from a height of eight (8) inches max. on top of the firing pin. The elapsed time between the ball striking the firing pin and the explosion or flash shall be measured to determine compliance with the requirement.

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4.4.3.2 Waterproofness.-The delay element shall be conditioned as specified in 3.4, then removed and subjected to the functioning test as specified in 4.4.3.1. Any delay element that fails to meet the requirement shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-Packaging shall be as specified on dwgs. 9258182 and 9258183.

5.2 Packing

5.2.1 Level C.-Packing shall be as specified on dwg. 9258184.

5.3 Marking.-Markings of all packaging and packing shall comply with dwgs. 9258182, 9258183 and 9258184.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval.

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6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (See 6.3.1 c.). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423).

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6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey, 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Check Test for Possible Deterioration.-The following test is designed to prevent major ammunition items being loaded with delay elements that may have deteriorated. This test shall be performed on delay elements from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the delay elements are loaded does not require a check test, the contracting officer shall insert into the contract a check test performance provision. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the deterioration check test.

6.4.1 Functioning (See 3.3).-The sample plan for this test shall be in accordance with MIL-STD-105 as listed below. Both "duds" and "delay time" shall be determined from the same sample. Any delay element which fails to comply with the specified requirements shall be classed defective. The test shall be performed in accordance with 4.4.3.1, using equipment in accordance with 4.3.4.

(1) Duds - Code Letter "L", Single Plan, AQL of 0.15%

(2) Delay Time - Code Letter "L", Single Plan, AQL of 0.65%

Custodian:
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Preparing Activity:
Army-MU

Project Number: 1375-A-019

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